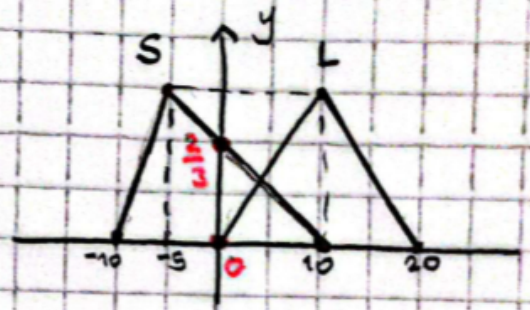
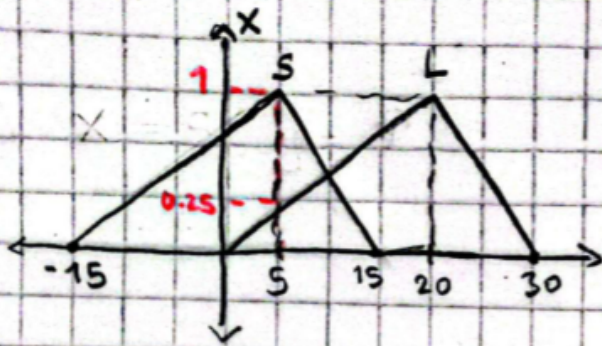
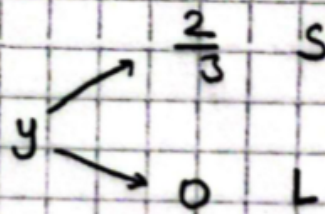
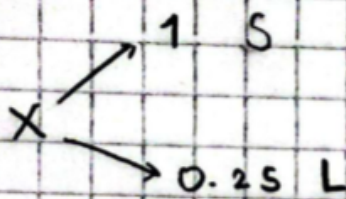




a-) $X = 5$, $y = 0$ Then :



$$b-) w_1 = \min(1, \frac{2}{3}) = \frac{2}{3} \quad z_1 = -5 + 4 \times 0 + 1 = -4$$

$$w_2 = \min(1, 0) = 0 \quad z_2 = -2 \times 0 + 1 = 1$$

$$w_3 = \min(0.25, \frac{2}{3}) = 0.25 \quad z_3 = -6 \times 5 + 7 = -23$$

$$w_4 = \min(0.25, 0) = 0 \quad z_4 = 5 + 3 \times 0 + 1 = 6$$

$$\text{Output} = \frac{\sum_{i=1}^n w_i \times z_i}{\sum_{i=1}^n w_i} = \frac{\frac{2}{3} \times (-4) + 0 \times 1 + \frac{1}{4} \times (-23) + 0 \times 6}{\frac{2}{3} + 0 + \frac{1}{4} + 0}$$

$$= \frac{-\frac{101}{12}}{\frac{11}{12}} \approx -9.18 //$$

$$C-) \quad w_1 = 1 \times \frac{2}{3} = \frac{2}{3}$$

$$z_1 = -5 + 4 \times 0 + 1 = -4$$

$$w_2 = 1 \times 0 = 0$$

$$z_2 = -2 \times 0 + 1 = 1$$

$$w_3 = 0.25 \times \frac{2}{3} = \frac{1}{6}$$

$$z_3 = -6 \times 5 + 7 = -23$$

$$w_4 = 0.25 \times 0 = 0$$

$$z_4 = 5 + 3 \times 0 + 1 = 6$$

$$\text{Output} = \frac{\sum_{i=1}^4 w_i \times z_i}{\sum_{i=1}^4 w_i} = \frac{-4 \times \frac{2}{3} + 1 \times 0 - 23 \times \frac{1}{6} + 6 \times 0}{\frac{2}{3} + 0 + \frac{1}{6} + 0}$$

$$= \frac{-\frac{39}{6}}{\frac{5}{6}} \approx -7.8 //$$